Engineering Mathematics - I · Unit_02 · Unit 2

DRAFT — UE26MA141A · Unit_02 · Unit 2 · Rules for finding the Complementary Function, Problems

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
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M1

Rules for finding the Complementary Function, Problems

1 Rules for finding the Complementary Function, Problems

Study focus: Rules for finding the Complementary Function, Problems. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000026 · EV0000027

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

From now on, we will be discussing about only higher order linear differential equations.

Key details

- We will be solving a higher.
- Order homogeneous linear differential equation with constant coefficients.
- We know that the general solution of homogeneous.
- LD with constant coefficients is called complementary function.
- Every homogeneous linear differential equation with constant coefficients has an algebraic equation associated with it.
- The algebraic equation is called auxiliary equation, and the general solution of the.
- Homogeneous linear differential equation depends on the nature of the roots of the auxiliary equation.
- The auxiliary equation has real and different rules, real and distinct rules.
- When the rules of the auxiliary equation are real and equal.
- And when the rules of the auxiliary equation are imaginary.
- Differentiating this y one time and two times with respect to x, we get y dash is equal to em into e power mx and y double dash is m square into e to the power mx.
- You can substitute, you can plug in these equations into the differential equation.
- Only when y is equal to e to the power mx is a solution of the differential equation.
- We know that a to the power mx is not equal to 0 for any real value of x.
- So, which means that m square plus k1m plus k2 equal to 0.

- K1 into capital D plus K2 of Y is equal to 0.
- Or how are we writing the auxiliary equation of the differential equation?
- This equation is f of b.
- of y is equal to 0.
- In f of d you just replace b by m.
- And equate the algebraic expression to 0, we get the auxiliary equation.
- Not a homogeneous differential equation of this type.
- The auxiliary polynomial of this differential equation is f of m.
- What is f of m? f of d is d squared plus k1 into capital D plus K2.
- In F of D replace D by M.
- Linear differential operator with constant coefficient f of d has.
- An auxiliary polynomial associated with it, which is F of M.
- Because the general solution of the homogeneous linear differential equation depends on the nature of the loads of the auxiliary equation.

Example or application from the lecture

- First let us learn how in detail about auxiliary equation.
- For that let us consider a simple second order homogeneous linear differential equation of the type d2 by b dx2 plus k1 d by dx plus k2 of y is equal to 0.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 05:00–09:59

If we are discussing three different cases.

Key details

- So what is the first case when the rules of the auxiliary equation m1 and m2 are real and distinct.
- So because the auxiliary equation is.
- A quadratic equation, it has two roots, m1 and m2.
- Y is equal to some constant c1 into e to the power m1x plus c2 into e to the power m2x.
- The two real and distinct roots of the auxiliary equation.
- What is x?
- X is the independent.
- variable C1 C2 are the arbitrary constant.
- This is how the general solution of the differential equation look like when the roots are real and distinct.
- See what is the order of the differential equation to.
- We have learned in the previous sessions that the general solution of a first order linear diffusion differential equation has one arbitrar constant in its general solution.
- Two arbitrary constants and an n th order differential equation has n arbitrary constants in its general solution because here the order is 2, there are two arbitrary constants in the solution.

- are real and distinct.
- So now first you need to write the auxiliary equation of this differential equation.
- In f of d replace d by m and equate it to 0.
- Another simple way of learning to write the auxiliary equation.
- In this differential equation, you replace y by 1, dy by dx by u.
- D2 y by dx2 by m square.
- If the differential equation has d3 y by dx3, replace it by m square.
- Solve this auxiliary equation, find out the linear factors and find the roots.
- The roots are minus 3 and 1.
- Analyze the roots.
- The roots are real and they are different.
- It is c1 e to the power m1x plus c to e to the power m2x.
- 2x plus c2 e to the power 1 in 2x.
- When the roots of the auxiliary equation are real and equal.
- M1 is equal to M2 is equal to M C.
- Then the general solution, the differential equation is y is equal to c1 plus c2 into x, the independent variable x into e power mx.
- The general solution of the homogeneous linear differential equation of order 2 if the roots of the auxiliary equation are real and equal.
- Y is equal to C1 plus C2x into E2.
- Yes, LDE with constant coefficient.
- And the auxiliary equation is m square plus 4m plus 4 is equal to 0, which is m plus 2 the whole square.

Example or application from the lecture

- If the roots of the auxiliary equation are real and distinct, then the general solution of this second order homogeneous linear differential equation is given by.
- Which is associated with the given homogeneous linear differential equation with constant coefficients.
- So let us illustrate this concept by taking an example.

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EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 10:00–14:59

And minus 2.

Key details

- The two rules are real and repeated.
- It is c1 plus c2 into x into e to the power em x.
- What is m over here?
- X is the solution of this homogeneous differential equation.
- Third case is when the roots are complex imaginary.
- Let the roots be m1 is equal to alpha plus i beta, m2 is alpha minus i beta.

- We know that the complex roots always occur in pairs, conjugate roots occur in pairs.
- complex is y is equal to e to the power alpha is c1 cos beta x plus c2 sine beta is understand this it is e to the power alpha is the real part of the root and beta is the imaginary part of the root so it is e to the alpha into independent variable x.
- into C1 arbitrar constant C1 cos beta x plus c2 sine beta s.
- equation are complex in nature.
- That is the auxiliary equation.
- Using the quadratic formula, find the roots of this equation.
- It is minus b plus r minus square root of b square minus 4ac by 2 into a.
- A is 1, B is minus 4, and C is 13.
- Substitute in the quadratic formula.
- Get the roots.
- The roots are 2 plus 3i, 2 minus 3i.
- Alpha is 2 and beta is 2.
- Because the roots of the auxiliary equation are complex.
- C1 cos 3x plus c2 sin 3x.
- This is the general solution when the roots are complex in nature.
- To an n thirder homogeneous linear differential equation.
- So this is the auxiliary equation because the degree of this algebraic equation is n.
- How many roots?
- The general solution is just extend the same concept which we learned for a second order homogeneous differential equation to an nth order differential equation.
- Plus C and E to the power mx.
- What is it when the roots m1, m2?
- M3 M and all are real and equal.
- Say equal to M1 equal to M2 equal to M3 equal to M and equal to M.
- be C1 plus C2X plus C3 X square plus C n x to the power n minus 1 into e to the power m the equal root mx.

Example or application from the lecture

- Now let us consider an example d2y by dx2 minus 4 into dy by dx plus 13y is equal to 0.
- Let us consider different possibilities.
- Let us consider.

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EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 15:00–19:59

You are stopping at x to the power n minus 1.

Key details

- If the order of the differential equation is 3, say you are stopping at c3 x to the power 3 minus 1, which is x to the power 2.

- If the order of the differential equation is 2, you are stopping at C2 x to the power 2 minus 1, which is x.
- Look at the number of arbitrary constants in the general solution.
- How many arbitrary constant in an nth order differential equation?
- See, there are n constant.
- C1, C2, C3, C.
- And the remaining n minus 4 roots are real and distinct.
- So the first four roots are real and equal to same m.
- Remaining n minus 4 are real and distinct.
- What is it?
- If there are two pairs of imaginary roots, alpha plus r minus i beta, gamma plus r minus i delta, and the remaining n minus 4 roots are real and distinct.
- See there are totally n roots.
- The first two roots are alpha plus i beta and alpha minus pi beta.
- The next two roots are gamma plus i delta, gamma minus i delta.
- The remaining n minus 4 are real and distinct.
- What is it?
- E to the power alpha x c1 cos beta s plus c2 sin beta x.
- Complex roots e to the power gamma x into c3 cos delta x plus c4 sine delta x and for the remaining n minus 4 roots it is c phi e to the power m phi x.
- Plus C6 e to the power mc plus c n e to the power mnx.
- Here I should write the constants as C3 and C4.
- It is written as C and C2.
- We have to change this.
- What is the other possibility?
- And the remaining n minus 4 roots say they are real and distinct.
- into C1 plus C2X for the repeated complex load C1 plus C2X into cos beta x plus c three plus c4 x into sine beta x.
- into for the remaining n minus four roots it is c phi e to the power m phi x plus c six e to the power m c plus cn e to the power m n x.
- If the complex roots are repeated in the auxiliary equation, then we write this way C1 plus C2x into cos beta x plus c3 plus c4 x into sin beta x.

Example or application from the lecture

- Let us say the first four roots, m1, m2, m1.
- differential equation see for the first four equal roots let us write it is c1 plus c2x plus c3x square plus c 4x to the power 3 because the first four roots are real and equal into e to the power mx.
- Let us discuss one more possibility.

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EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

Core explanation · 20:00–24:59

Homogeneous linear differential equation with constant coefficients.

Key details

- A third order homogeneous linear differential equation.
- Dq y by dx q plus 4 d by dx is equal to zero.
- What is it?
- It's a homogeneous third order LT with constant coefficients.
- General solution of this differential equation called complementary function.
- What is the auxiliary equation?
- F of D equal to when equate.
- f of m to zero.
- See this linear differential operator.
- For every linear differential operator, we can introduce a characteristic polynomial mq plus 4m and the.
- And when mq plus 4m is equated to 0, we get the auxiliary equation.
- I have taken a simple cubic equation.
- 3 roots, because the degree of this equation is 3.
- to zero m is equal to zero and m is equal to plus or minus divide.
- Here C 0 for the real root I'm writing C1 E to the power 0x.
- sine to x does the solution of this c one plus c2 cos to x plus c three sin to x.

Example or application from the lecture

- Let us solve.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Rules for finding the Complementary Function, Problems transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/002_Rules_for_finding_the_Complementary_Function_Problems__4a39dee530243163.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Rules for finding the Complementary Function, Problems transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/002_Rules_for_finding_the_Complementary_Function_Problems__4a39dee530243163.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Rules for finding the Complementary Function, Problems transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/002_Rules_for_finding_the_Complementary_Function_Problems__4a39dee530243163.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9989; only the JSON segmentation supplies the displayed timestamps.
Rules for finding the Complementary Function, Problems transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/002_Rules_for_finding_the_Complementary_Function_Problems__4a39dee530243163.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.

3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00006	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00007	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...
SRC00008	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00009	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...

C Revision and retrieval practice